Exploring Lightweight Structures for Tiny Object Detection in Remote Sensing Images

Dongyang Liu, *Student Member, IEEE*, Junping Zhang, *Senior Member, IEEE*, Yunxiao Qi, *Student Member, IEEE*, Yunqiao Xi, *Student Member, IEEE*, and Jing Jin, *Member, IEEE*

***Abstract*— Detecting tiny objects in remote sensing images has been an intriguing yet challenging topic in remote sensing image processing. While significant progress has been made in many studies, most existing methods focus on improving the accuracy of tiny object detection without particular consideration for computational complexity, which restricts their applicability in resource-limited conditions. Therefore, this article aims to design a lightweight detection algorithm tailored for tiny objects in remote sensing images. First, we investigate the impact of the complexity of different components in deep learning-based object detection models on the accuracy of tiny object detection, including the backbone and detection head. Then, a dedicated backbone for tiny object detection is proposed, achieving competitive detection accuracy while remaining lightweight. Moreover, we propose a lightweight detection head that incorporates deformable convolution and optimize the channel dimension. Finally, we combine the above methods to introduce a lightweight network, lightweight tiny object detecection network (LTDNet), for tiny object detection in remote sensing images. Benefiting from the dedicated designs for the backbone and detection head specifically for tiny objects, the proposed method can achieve competitive detection accuracy with very low parameters and computational complexity. Extensive experiments are conducted on the artificial intelligence (AI)-TODv2 and learning vision and remote sensing (LEVIR)-Ship datasets, and the results demonstrate the effectiveness of our proposed method. Specifically, the proposed method achieves 54.6% AP50 on the AI-TODv2 dataset with only 4.85 M parameters and 38.19 G floating-point operations (FLOPs). The code will be released soon on the GitHub repository: https://github.com/dyl96/LTDNet**

***Index Terms*— Feature extraction backbone, lightweight, remote sensing images, tiny object detection.**

I. Introduction

REMOTE sensing images have been widely used in numerous tasks, such as change detection [1], [2], land cover classification [3], clustering [4], [5], and object detection. Remote sensing image object detection (RSOD) tasks have numerous applications in both military and civilian domains, including military reconnaissance, traffic monitoring, and maritime rescue operations. Currently, RSOD has become a highly researched topic due to its wide-ranging applications and implications [6], [7], [8], [9], [10], [11], [12].

The object detection methods can be divided into two categories: traditional object detection methods and deep learning-based methods. Traditional approaches to object detection often rely on manually designed features [13], [14]. However, the intricacies of backgrounds and the variability of objects pose challenges, rendering traditional methods inadequate due to their limited capacity for representation [15], [16]. Recently, the advancement of deep learning, particularly convolutional neural networks (CNNs), has presented a novel prospect for RSOD. Unlike conventional methods, deep learning-based techniques can automatically learn useful features, thereby improving detection performance. Several studies have incorporated deep learning techniques into RSOD [9], [10], [17], [18]. These methods focus on various aspects, such as rotation invariance, accurate object localization in high-resolution images, integration of contextual information, and adaptive attention fusion mechanisms. These methods aim to improve the effectiveness and robustness of RSOD and have yielded promising results.

However, unlike natural images, remote sensing images often contain numerous tiny objects, as shown in Fig. 1. Despite the significant progress achieved in the field of object detection with the development of deep learning technologies, most existing object detectors are primarily designed for objects of standard sizes. The limited detail information of tiny objects in remote sensing images compared to larger objects makes feature extraction and utilization more challenging. This difficulty leads to more challenging localization and classification of tiny objects, often resulting in suboptimal performance. Therefore, numerous methods have been developed to address the challenge of detecting tiny objects in remote sensing images.

Several methods enhance the accuracy of tiny object detection in remote sensing images by improving sample allocation strategies [20], [21], [22], [23]. In addition, there are some methods to improve the performance of tiny object detection from the perspective of feature enhancement, such as [24], [25], [26], and [27]. These methods have shown promising performance.

However, most of these methods focus on improving accuracy without particular optimization in terms of algorithm

Received 30 December 2024; revised 21 February 2025 and 26 April 2025; accepted 2 May 2025. Date of publication 6 May 2025; date of current version 19 May 2025. This work was supported by the National Natural Science Foundation of China under Grant 62271171. *(Corresponding author: Junping Zhang.)*

Dongyang Liu, Junping Zhang, Yunxiao Qi, and Yunqiao Xi are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: 20B905060@stu.hit.edu.cn; zhangjp@hit.edu.cn).

Jing Jin is with the Department of Control Science and Engineering, Harbin Institute of Technology, Harbin 150001, China.

Digital Object Identifier 10.1109/TGRS.2025.3567345

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.



Fig. 1. Examples of tiny objects in remote sensing images from the AI-TODv2 dataset 0. These objects are very small in size and their appearance features are not obvious.

complexity. This limitation significantly restricts their applicability in remote sensing applications where algorithm complexity is critical. Moreover, existing methods mostly improve upon generic object detection frameworks such as Faster R-CNN [28], fully convolutional one-stage object detection (FCOS) [29], and RetinaNet [30] by directly utilizing backbone, neck, and head components without specific modifications tailored for tiny object detection. This raises the question: are existing generic object detection structures truly suitable for detecting tiny objects in remote sensing images? Is a backbone network like ResNet50 [31] indeed the best choice for feature extraction in tiny object detection?

With these questions in mind, this article aims to explore a remote sensing tiny object detection model with low complexity and competitive accuracy. The research focuses on two key aspects to optimize the model: structural efficiency and rationality. Specifically, we explore the backbone and head design for tiny object detection and ultimately develop a lightweight structural framework for this task.

Our main contributions are summarized as follows.

1) We propose a lightweight backbone specifically designed for tiny object detection in remote sensing images. Recognizing the importance of localization accuracy and detailed information for detecting tiny objects, the proposed backbone shifts computational resources from the later stages to the first two stages. This strategy enhances the extraction of fine-grained features. In addition, in order to maintain the lightweight of the backbone, the advanced lightweight block is adopted as the basic unit.
2) We propose a lightweight detection head for tiny objects. Considering the importance of dynamic receptive fields in tiny object detection, we incorporate a deformable convolutional network (DCN) to enhance feature adaptability. Furthermore, in order to maintain the lightweight of the detection head, we further optimize its structure and parameters, achieving better results with lower complexity.
3) We propose a dedicated model specifically designed for tiny object detection in remote sensing images. This model integrates a specialized backbone and detection head, ensuring high detection accuracy while maintaining extremely low complexity. Extensive experiments on the artificial intelligence (AI)-TODv2 and learning vision and remote sensing (LEVIR)-Ship datasets demonstrate the effectiveness of the proposed method.

The rest of this article is organized as follows. The related work is presented in Section II. The details of the proposed lightweight tiny object detecection network (LTDNet) are described in Section III. The experimental setup and results are shown in Section IV. Finally, the conclusion is given in Section V.

II. Related Work

A. Generic Object Detection Methods

Many generic object detection methods have been proposed in recent years. Early methods relied on handcrafted features extracted from images. However, these methods only extract shallow features and have limitations in obtaining deep semantic information. With the development of deep learning technology, many deep learning-based methods have been proposed to overcome the above problems. The deep learning-based generic object detection methods can be divided into three main categories: single-stage, two-stage, and detection with Transformers (DETR)-like methods [32].

Single-stage object detection methods directly extract features from the input image and predict both the location and class. Specifically, these methods can be classified into anchor-based and anchor-free methods. The anchor-based methods, such as [38], [39], [40], [41], and [42], typically involve predefined anchor boxes of various sizes on feature maps. During training, these anchors are adjusted to align with ground truth object boxes, while object classes are determined through a classification branch. However, the fixed anchor sizes may not be practical for objects of varying sizes in real-world scenarios. As a result, anchor-free methods have been developed [43], [44], [45], [46]. These methods abandon the predefined anchor pattern and model the object bounding boxes regression through various approaches, such as keypoint regression, object center point regression, width and height regression, and determining the position by a set of points.

Unlike single-stage methods, two-stage object detection methods first extract the candidate regions that may contain objects from the input image and then classify and regress bounding boxes for these regions through extra one or more stage. This process is more complex compared to single-stage methods. Common two-stage object detection methods include [47], [48], [49], [50], and [51].

Different from the above methods, DETR proposes an end-to-end object detection paradigm. DETR adopts a backbone to extract the image features and Transformer encoder to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

generate high-dimensional feature representations. Finally, the Transformer decoder uses learnable queries to predict the objects. Despite achieving promising results, DETR faces issues such as high computational complexity, slow convergence speed, and long training time. Therefore, many subsequent methods have been proposed to address these issues [33], [34], [35], [36], [37]. For instance, deformable DETR [33] introduces a deformable attention module to reduce computational complexity. Conditional DETR [34] accelerates training convergence speed through the use of a conditional decoder. Accelerate DETR training by introducing query denoising (DN-DETR) [35] introduces a denoising training mechanism to accelerate model convergence and improve detection performance. Dynamic anchor box DETR (DAB-DETR) [36] reintroduces anchor boxes into DETR to accelerate the convergence.

Although these methods perform well for larger objects in natural or remote sensing images, they lack specialized designs for tiny objects. Thus, directly applying them to tiny object detection often fails to achieve satisfactory performance.

B. Tiny Object Detection Methods

To enhance the accuracy of tiny object detection, numerous methods have been successively proposed in recent years. In summary, these methods can mainly be categorized into three kinds: data augmentation, feature enhancement, and improved sample assignment strategies.

1) Data Augmentation: This category of methods improves tiny object detection by increasing the quantity of tiny objects. To enhance spatial diversity and improve the alignment between detection boxes and ground truth boxes, Kisantal et al. [52] suggested augmenting the training data by duplicating and inserting tiny objects multiple times across the image. Moreover, Chen et al. [53] introduced a context-aware adaptive resampling technique for data augmentation, which builds upon the aforementioned methods. This technique employed copy and paste operations while considering contextual information to mitigate issues like scale or background mismatches.

2) Feature Enhancement: Tiny objects often exhibit weak features due to their small size. Consequently, various methods have been proposed to enhance the features of tiny objects for improved detection accuracy. Among these, feature fusion stands out as a common approach, which fuses the semantic information from deep layers with the detailed information from shallow layers. Feature pyramid network (FPN) [19], for instance, merges adjacent features through both bottom–up and top–down pathways to enhance the features. Subsequent advancements, such as path aggregation feature pyramid network (PAFPN) and bi-directional feature pyramid network (BiFPN) [54], [55], [56], have further refined this fusion process. In addition, there are some methods to enhance the features of tiny objects through other means. For instance, Haris et al. [57] integrated image super-resolution with object detection for end-to-end joint training, thereby enhancing the features of tiny objects. Li et al. [58] and Bai et al. [59], on the other hand, introduced generative adversarial networks into tiny object detection, leveraging adversarial learning between the generator and discriminator to generate high-resolution feature representations, thereby enhancing detection performance. Denoising feature pyramid network with trans R-CNN (DNTR) [25] introduced denoising FPN and self-attention to suppress noise in the feature map and focus on the feature representation of tiny objects. Object reconstruction feature enhancement network (ORFENet) [26] addressed tiny object information loss in deep networks by incorporating an object reconstruction branch and introduced a multiscale adaptive feature enhancement module to dynamically adjust the features of different receptive fields. Tiny object detection network based on explicit semantic guidance (ESG_TODNet) [27] employed explicit semantic guidance operations to improve the features of tiny objects, ultimately enhancing the detection accuracy.

3) Improved Sample Assignment Strategy: The sample assignment method is a crucial aspect of deep learning-based object detection methods. Classic approaches employ an effective positive and negative sample assignment strategy based on intersection over union (IoU) thresholds. However, such methods may generate numerous low-quality samples, making them less effective for detecting tiny objects. Therefore, some methods aim to enhance the detection performance of tiny objects by improving the sample assignment strategy. For example, adaptive training sample selection (ATSS) [60] automatically selected suitable positive and negative samples by analyzing the statistical characteristics of the objects. Xu et al. [20] proposed a metric called "Dot Distance" to improve the selection process of positive and negative samples. On the other hand, Xu et al. [22] enhanced the anchor-based tiny object detector by improving the measurement method between the ground truth boxes and prior boxes. They replace the original IoU measurement method with the Gaussian distribution-based Wasserstein distance, thereby improving the quality and quantity of positive samples. M-Center [21] improved tiny object detection performance by considering multiple center points. Meanwhile, receptive field-based label assignment [23] modeled prior boxes or points based on receptive fields and used the receptive field distance to measure the distance between ground truth boxes and prior boxes.

However, while these methods have achieved certain effectiveness, most of them primarily focus on improving accuracy without considering complexity. This limits these methods' direct applicability to scenarios where algorithm complexity is a critical concern. Therefore, this article addresses the issue by focusing on reducing the complexity of the tiny object detection models for remote sensing images.

C. Lightweight Model

Lightweight networks offer an effective means to reduce the complexity of deep learning-based algorithms. Currently, numerous lightweight network techniques have been proposed, which can be categorized into two main types: CNN architectures and Transformer architectures.

Lightweight architectures based on CNN primarily involve improving classic convolution operations, including

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

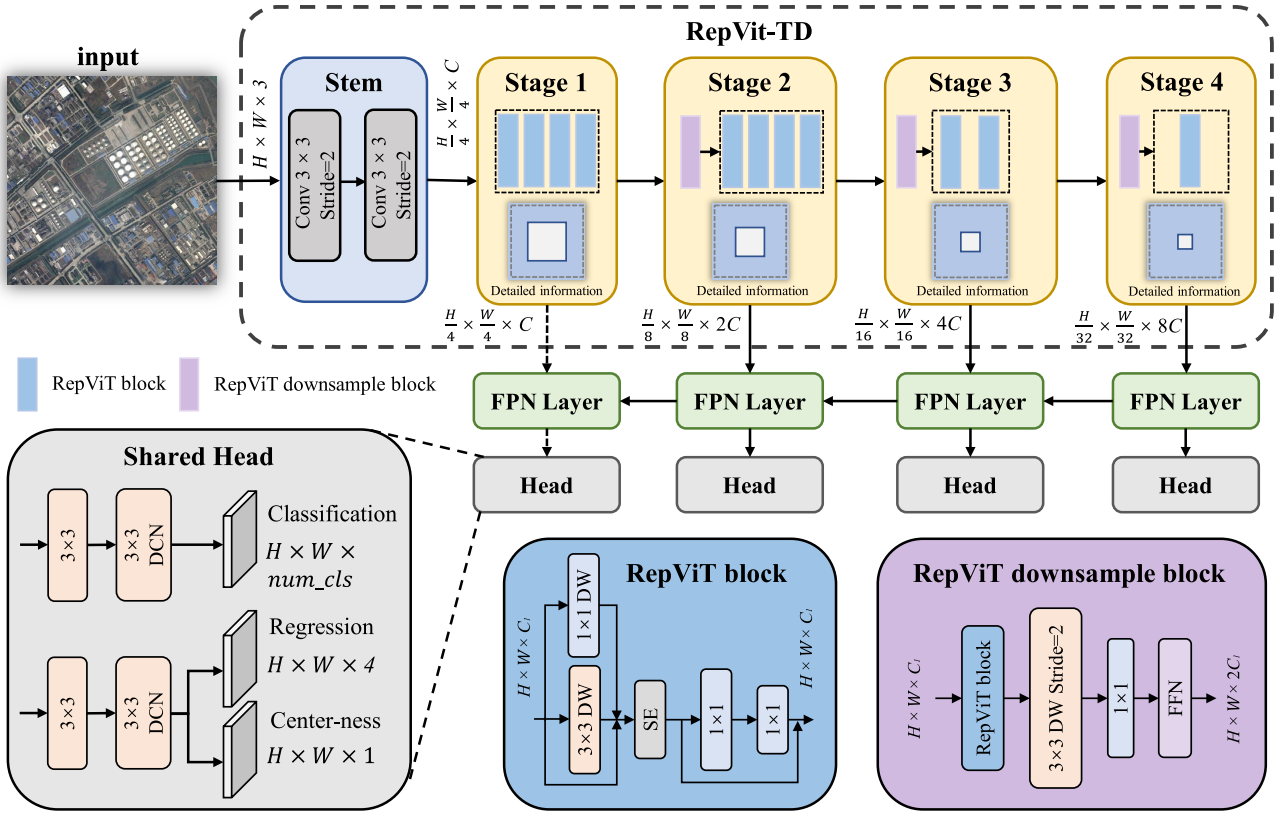


Fig. 2. Overall structure of the proposed LTDNet. The lightweight network LTDNet proposed in this article is designed for detecting tiny objects in remote sensing images. Its backbone consists of a novel structure specifically designed for tiny object detection, which is constructed using RepViT block and RepViT downsample block as basic components. The computational complexity at different stages is optimized to adapt to the detection of tiny objects. For feature fusion, the FPN structure is employed. In designing the detection head, a tradeoff between accuracy and complexity is considered, and a combination of Conv2d and DCN is adopted.

depth-wise separable convolution [61], inverted residual bottlenecks [62], channel shuffling [65], [66], and structural reparameterization [63]. These methods have paved the way for the development of several widely used lightweight CNN architectures. For example, the MobileNets series [61], [62], [64] mainly construct lightweight models by utilizing depth-wise separable convolutions, ShuffleNets [65], [66] rely on channel shuffling, EfficientNet [67] utilizes compound scaling for efficient model design, and reparameterization convolutions in a ViT-like metaformer structure (RepViT) [68] is constructed using structure reparameterization techniques.

Lightweight architectures based on Transformer are designed by leveraging the Transformer structure. These architectures mainly combine Transformer with CNN and improve the original multihead self-attention (MHSA) component to reduce the complexity. For example, MobileViT [69] adopted a hybrid architecture, combining lightweight MobileNet blocks with MHSA blocks. EfficientFormer [70] introduced a dimension-consistent design paradigm to enhance latency performance boundaries. EfficientFormerv2 [71] is a further optimized version of EfficientFormer, enhancing the performance.

However, although these models have relatively lightweight characteristics, most of them are universal backbone for image classification, object detection, and image segmentation applications. This is not entirely applicable for tiny object detection in remote sensing images. Therefore, this article designs a dedicated lightweight backbone for tiny object detection in remote sensing images.

III. Proposed Method

This article explores a lightweight model structure for detecting tiny objects in remote sensing images, thus avoiding the higher complexity. The proposed method is built upon the anchor-free method FCOS that commonly used in general object detection. The overall architecture of the proposed LTDNet is illustrated in Fig. 2. The proposed LTDNet follows the basic paradigm of FCOS. It consists of three main components: a backbone for feature extraction, a neck for feature fusion, and a head for object prediction.

The basic structure of FCOS, the design of the lightweight backbone, and the lightweight head will be introduced in detail.

A. Basic Structure of FCOS

FCOS is a universal object detection method based on anchor-free techniques, comprising three main components: the backbone network, the feature fusion network, and the detection head. Given an input image size of $H \times W$, the backbone network extracts features of different scales. Subsequently, these features undergo a feature fusion network to generate fused features. Finally, these features are fed into the detection head for object classification and localization. In the standard FCOS implementation, the backbone is ResNet50, the feature fusion network is FPN, and the detection head consists of classification branches along with regression and centerness branches, each containing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

TABLE I
FLOPS AND PARAMS IN DIFFERENT PARTS OF STANDARD FCOS, AS WELL AS THE RATIO OF DIFFERENT PARTS. FLOPS REPRESENT FLOATING-POINT OPERATIONS, AND PARAMS REPRESENT THE NUMBER OF PARAMETERS

	Total	Backbone	Neck	Head
FLOPs(G)	122.89	49.31	10.17	63.42
ratio(%)	100	40.13	8.28	51.61
Params(M)	31.85	23.23	3.87	4.75
ratio(%)	100	72.94	12.15	14.92

four layers of standard 3×3 convolutions. The standard FCOS implementation starts from the P3 feature map whose size is $H/4 \times W/4$. However, for tiny objects, higher resolution feature maps can provide better spatial information and more refined features, which are crucial for tiny object detection. Therefore, this article not only discusses and studies the standard FCOS but also conducts experiments on the structure that incorporates higher resolution feature maps.

The floating-point operations (FLOPs) and the number of parameters (Params) of different parts of FCOS are shown in Table I. From the table, it can be observed that the backbone and head contribute the most to the computational load, with Params accounting for 72.94% and 14.92%, respectively, and FLOPs accounting for 40.13% and 51.61%, respectively. Therefore, the optimization of this article is also carried out from the backbone and detection head.

B. Lightweight Backbone for Tiny Object Detection

In this section, we explore the influence of the backbone network on tiny object detection in remote sensing images, and the results are presented in Tables II and III. We have conducted experiments using both common-sized backbones and lightweight backbones. From Table II, it can be seen that the backbone network has a significant impact on the performance, and different backbones lead to considerable performance differences. For instance, FCOS with ResNet50 as the backbone achieves an average precision (AP) of 15.0%. As the scale decreases, networks with ResNet34 and ResNet18 as the backbone exhibit a serious accuracy decrease. However, FCOS with MobileNetv2 and ShuffleNetv2 as the backbone can achieve AP of 9.11% and 8.13%, respectively. Despite these backbones have lower Params and FLOPs, the detection accuracy of tiny objects of MobileNetv2 and ShuffleNetv2 is significantly compromised, which is unacceptable considering the already low detection accuracy. The experimental results in Table III are similar to those in Table II. Therefore, designing a lightweight backbone specifically tailored for tiny object detection with satisfactory performance is crucial. The RepViT backbone, with more advanced structural design, maintains relatively good detection accuracy under lightweight conditions. It achieves better performance compared to ResNet34 while having fewer Params and FLOPs. However, there is still a considerable gap compared to the ResNet50 backbone.

Through the observation of the design of different backbones, we can draw some conclusions. Most of these backbones are primarily designed for image classification. The computational load assigned to feature maps at different scales mostly follows a pattern of fewer blocks at higher resolutions. The comparison of different backbones and their distribution of computational blocks across stages is presented in Table IV. For example, ResNet50 has three, four, six, and three blocks in its four stages; MobileNetV2 has three, four, six, and two blocks; MobileNetV3 has one, two, five, and two blocks; EfficientNet has two, three, seven, and one blocks; and EfficientFormerV2 has three, three, nine, and nine blocks, while the RepViT backbone allocates two, two, 14, and two blocks across its four stages. It can be observed that these backbones allocate relatively low computational load to the first two stages, as more blocks would result in significant computational overhead at higher resolutions. However, the detection of tiny objects in remote sensing images usually requires high positioning requirements, and the detailed information of these tiny objects is relatively limited. The earlier stages (stages 1 and 2) operate on high-resolution feature maps and are more critical for preserving spatial details and localization precision. These aspects are especially important in tiny object detection, where the objects often occupy only a few pixels and fine-grained spatial cues play a vital role. If these details are not taken seriously, it will inevitably affect the detection accuracy of tiny object. Moreover, from the perspective of lightweight, more attention should be paid to the distribution of computational resources across feature maps of different sizes in the backbone to ensure that the allocation is both effective and beneficial for tiny object detection. Therefore, designing a dedicated lightweight backbone for tiny object detection becomes essential. Inspired by [68] and [72], a lightweight backbone specifically designed for tiny object detection is proposed, based on RepViT block and named RepViT-TD. The details of RepViT-TD are illustrated in Fig. 2. We aim to design a backbone network that matches the accuracy of ResNet50 while being significantly lightweight.

The design of RepViT-TD follows the principles of lightweight and is suitable for tiny object detection in remote sensing images. For the stem section, RepViT-TD only uses two layers of standard 3×3 convolution with a stride of 2. For different stages of feature extraction, we redesign the proportion of computation for different stages. Considering that detailed information is particularly important for the detection of tiny objects, we adopted more computation in feature maps with higher resolution at the bottom layer and set the number of RepViT blocks for the four stages to 4, 4, 2, and 1, respectively. Moreover, RepViT downsample blocks are adopted at the beginning of the last three stages. In addition, considering the issue of computational complexity, we adopted a relatively small channel in the four stages, which are 64, 128, 256, and 512.

The structures of RepViT block and RepViT downsample block are shown in Fig. 2. The RepViT block adopts the forms of spatial mixer and channel mixer, where the spatial mixer adopts a multibranch reparameterization structure and the channel mixer adopts a feedforward network (FFN) structure. In addition, an extremely low computational squeeze-and-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

TABLE II

IMPACT OF DIFFERENT BACKBONES ON THE PERFORMANCE OF TINY OBJECT DETECTION IN REMOTE SENSING IMAGES. ALL METHODS WERE TRAINED FOR 12 EPOCHS ON THE AI-TODV2 DATASET

Method	Backbone	Params(M)	FLOPs(G)	AP(%)	AP50(%)	AP75(%)	AP_{vt}(%)	AP_t(%)	AP_s(%)	AP_m(%)
FCOS	Common:									
	ResNet50	31.85	123.16	15.0	37.5	9.1	3.2	13.4	20.3	31.5
	ResNet34	28.97	118.74	11.8	30.8	6.7	2.8	10.9	16.4	24.9
	ResNet18	18.94	95.1	10.3	27.1	6.1	1.6	9.6	14.4	21.3
	Swin-T	34.39	129.97	13.3	34.7	7.3	2.9	12.2	18.1	27.7
	Lightweight:									
	MobileNetV2	9.11	74.54	9.8	26.4	5.2	1.6	8.5	14.6	21.4
	ShuffleNetV2	8.13	72.64	7.7	20.9	4.1	1.3	7.4	11.6	17.1
	EfficientFormerV2	13.5	79.91	11.0	29.1	5.8	2.1	10.2	15.0	24.0
	RepViT	12.56	82.20	13.5	34.8	7.9	3.0	12.5	18.8	28.1

TABLE III

IMPACT OF DIFFERENT BACKBONES ON THE PERFORMANCE OF TINY OBJECT DETECTION IN REMOTE SENSING IMAGES. ALL METHODS WERE TRAINED FOR 12 EPOCHS ON THE LEVIR-SHIP DATASET

Method	Backbone	Params(M)	FLOPs(G)	AP50(%)
FCOS	Common:			
	ResNet50	31.84	50.2	80.6
	ResNet34	28.95	48.46	76.8
	ResNet18	18.93	38.79	75.9
	Swin-T	34.37	53.68	78.6
	Lightweight:			
	MobileNetV2	9.05	30.38	68.6
	ShuffleNetV2	8.09	29.65	68.2
	EfficientFormerV2	13.49	32.54	73.7
	RepViT	12.54	33.53	79.0

TABLE IV

COMPUTATIONAL BLOCK DISTRIBUTION ACROSS STAGES FOR DIFFERENT BACKBONES

Backbone	Stage1	Stage2	Stage3	Stage4
ResNet50	3	4	6	3
MobileNetV2	3	4	6	2
MobileNetV3	1	2	5	2
ShuffleNetV2	1	4	8	4
EfficientNet	2	3	7	1
EfficientFormerV2	3	3	9	3
RepViT	2	2	14	2
RepViT-TD	4	4	2	1

excitation (SE) layer is added between the spatial mixer and the channel mixer. During inference, the multibranch structure of the spatial mixer can be merged into a 3 × 3 depth-wise convolution. The principle is shown in Fig. 3 and the following formula.

Let the 3 × 3 depth-wise convolution and the 1 × 1 depth-wise convolution be

$$\text{Conv}_3(x) = W_3(x) + b_3 \tag{1}$$

$$\text{Conv}_1(x) = W_1(x) + b_1. \tag{2}$$

The output y in the spatial mixer s can be represented as

$$\begin{aligned} y &= \text{Conv}_3(x) + \text{Conv}_1(x) + x \\ &= W_3(x) + b_3 + W_1(x) + b_1 + x \\ &= (W_3 + W_1 + 1)(x) + (b_3 + b_1). \end{aligned} \tag{3}$$

Merging the above equation yields

$$\begin{aligned} W_{\text{final}} &= W_3 + W_1 + 1 \\ b_{\text{final}} &= b_3 + b_1. \end{aligned} \tag{4}$$

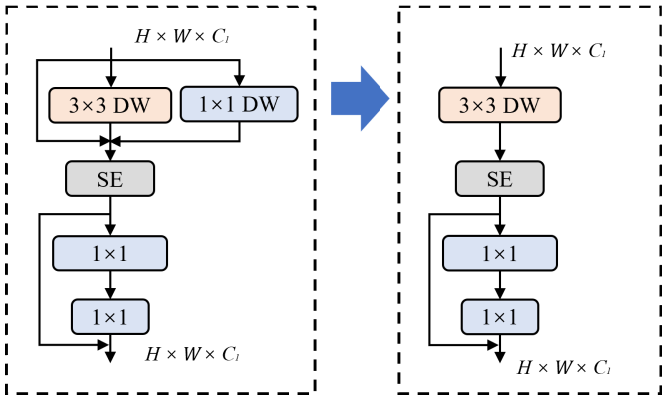


Fig. 3. Structure of RepViT block during training and inference. The left image shows the structure used during training, and the right image shows the structure used during inference.

Ultimately, it can be expressed as

$$y = W_{\text{final}}(x) + b_{\text{final}}. \tag{5}$$

Therefore, the three-branch structure used in the spatial mixer during training can be merged into a single 3 × 3 depth-wise convolution.

The RepViT downsample block adopts a relatively deep design. At the beginning, a RepViT block is used, followed by a 3 × 3 depth-wise convolution with a stride of 2 to downsample the spatial dimension. Finally, a 1 × 1 point-wise convolution and FFN are used for channel information mixing. Finally, the computational complexity of different parts of the proposed RepViT-TD is shown in Fig. 4.

C. Lightweight Head for Tiny Object Detection

In this section, we explore the detection head that has the greatest impact on the computational complexity, and the results are shown in Table V. First, we reduced the four convolutional layers in FCOS to three and two layers. From the table, it can be observed that after reducing some convolutional layers in the head, both FLOPs and Params decreased, while the detection accuracy for tiny object is almost unaffected. Thus, it can be inferred that achieving satisfactory results with only two layers of convolution in the head is feasible. Additionally, considering that tiny object detection may require a larger receptive field and taking

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

TABLE V

IMPACT OF DIFFERENT HEADS ON THE PERFORMANCE OF TINY OBJECT DETECTION. CONV2D REPRESENTS 2-D CONVOLUTION, DW REPRESENTS DEPTH-WISE CONVOLUTION WITH A KERNEL SIZE OF 5 × 5, AND DCN REPRESENTS DEFORMABLE CONVOLUTION NETWORK. ALL METHODS WERE TRAINED FOR 12 EPOCHS ON THE AI-TODV2 DATASET

Method	Head	Params(M)	FLOPs(G)	AP(%)	AP50(%)	AP75(%)	AP_{vt}(%)	AP_{t}(%)	AP_{s}(%)	AP_{m}(%)
FCOS	Conv2d × 4	31.85	123.16	15.0	37.5	9.1	3.2	13.4	20.3	31.5
	Conv2d × 3	30.67	107.41	15.4	39.3	9.2	3.2	13.7	21.0	31.4
	Conv2d × 2	29.49	91.39	15.2	39.7	8.6	3.2	13.6	20.4	30.7
	DW + Conv2d	28.46	77.58	12.6	33.2	6.8	2.7	11.0	17.7	28.4
	Conv2d + DCN	29.62	93.05	15.9	40.5	9.6	3.8	14.3	21.7	31.7
	DW+DCN	28.58	79.24	12.6	32.9	7.0	2.2	10.9	17.8	29.2

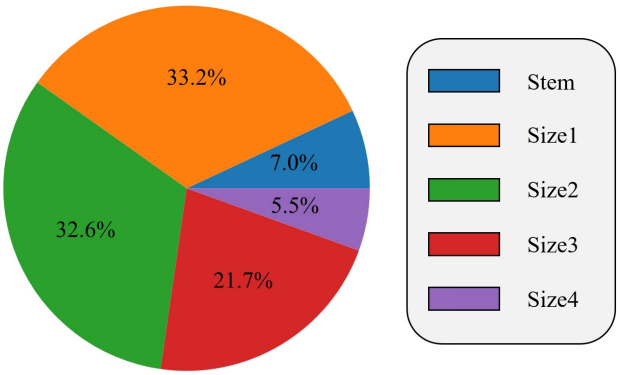


Fig. 4. Proportion of FLOPs for each part of the proposed RepViT-TD that is designed for tiny object detection of remote sensing images. Size1, Size2, Size3, and Size4 represent feature maps corresponding to 1/4, 1/8, 1/16, and 1/32 of the original image size, respectively. When the input image size is 800 × 800, the FLOPs of Size1 and Size2 with higher resolution feature maps account for 33.2% and 32.6%, respectively.

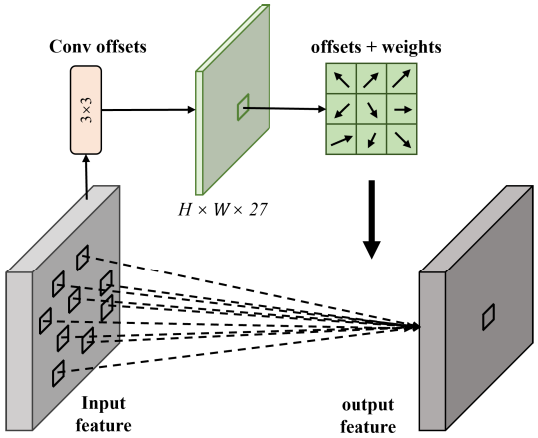


Fig. 5. Principle of DCN. First, use Conv offsets to calculate the offsets and weights of each point, and then, guide the calculation of convolution.

into account the lightweight of the model, we replaced one of the convolution layers in the head with a 5 × 5 depth-wise convolution. From the table, it can be seen that when the convolution layer is replaced with 5 × 5 depth-wise convolution, both FLOPs and Params decrease, but the detection accuracy significantly decreases. Furthermore, considering the importance of dynamic receptive fields for tiny object detection, we introduce a deformable convolution network (DCN) [73]. Although there is a slight increase in FLOPs and Params when replacing stand convolution with DCN, the increase in detection accuracy is substantial. Moreover, when two layers of convolution are replaced with 5 × 5 depth-wise convolution and DCN, both FLOPs and Params decrease, but the accuracy is lower. Therefore, the detection head design of LTDNet adopts the form of stand convolution and DCN, ensuring both accuracy and lower Params and FLOPs.

The principle of DCN is shown in Fig. 5. First, a conv-offsets is used to obtain the offset of each point in the x- and y-directions, as well as the weight of each point. Then, these offsets and weights guide the sampling positions and weights for convolution.

The calculation process of DCN can be expressed by the following formula:

$$y(p) = \sum_{k=1}^{K} w_k \cdot x(p + p_k + \Delta p_k) \cdot \Delta m_k \tag{6}$$

where $K = 9$, representing nine points of 3 × 3 convolution; w_k is the weight parameters of different points; p_k is the kth point among the region of 3 × 3; Δp_k denotes the offset learned by conv-offsets; and Δm_k indicates the weight coefficients learned by conv-offsets. This dynamic receptive field mechanism makes it easier for DCN to focus on critical regions and capture tiny objects.

D. Loss Function

The loss function of the proposed LTDNet includes three aspects: regression loss, classification loss, and centerness loss.

The regression loss Loss_{reg} can be formulated as

$$\text{Loss}_{\text{reg}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N_{\text{pos}}} L_{\text{DIOU}}\left(B_i, B_i^*\right) \tag{7}$$

where L_{DIOU} indicates the DIOU loss; N_{pos} denotes the number of positive samples; and B_i and B_i^* mean the predicted box and ground truth box, respectively.

The classification loss is computed using a focal loss. It can be defined as

$$\text{Loss}_{\text{cls}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N} \text{FL}\left(\boldsymbol{p}_i, \boldsymbol{y}_i^*\right) \tag{8}$$

where N denotes the number of total samples; $\boldsymbol{p}_i$ and $\boldsymbol{y}_i^*$ indicate the prediction and ground truth labels, respectively; and FL indicates the focal loss.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

The centerness loss is calculated using cross-entropy loss

$$\text{Loss}_{\text{cent}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N_{\text{pos}}} \text{CE}\left(\boldsymbol{c}_i, \boldsymbol{c}_i^*\right) \tag{9}$$

where $\boldsymbol{c}_i$ and $\boldsymbol{c}_i^*$ denote the prediction and ground truth, respectively.

The total loss can be summarized as

$$\text{Loss} = \text{Loss}_{\text{reg}} + \text{Loss}_{\text{cls}} + \text{Loss}_{\text{cent}}. \tag{10}$$

IV. Experiments

In this section, a detailed introduction to the dataset, evaluation metrics, and training settings used in the experiment will be provided. Then, an in-depth analysis of the experimental results will be conducted.

A. Datasets

This article utilizes three datasets, including ImageNet-1K [74] dataset for pretraining the proposed RepViT-TD, and two datasets AI-TODv2 [22] and LEVIR-Ship [75] for tiny object detection in remote sensing images.

1) ImageNet-1K: In order to provide a good initial parameter for the RepViT-TD backbone proposed in this article, we first conducted pretraining using the ImageNet-1K dataset. This dataset contains 1000 categories and the training data exceed 1 million.

2) Ai-TODv2: This dataset is used to evaluate the performance of tiny object detection in remote sensing images. It comprises eight categories and 28 036 remote sensing images. The object categories are: airplane, bridge, storage tank, ship, swimming pool, vehicle, person, and windmill. There are 752 745 object annotations. The image size in this dataset is 800 × 800 and is sourced from various publicly available remote sensing image datasets. Notably, all objects in this dataset are smaller than 64 × 64 pixels, with 86% of them being smaller than 16 × 16 pixels. The training set, validation set, and test set contain 2/5, 1/10, and 1/2 of the total number of images, respectively. In this study, we combine the training and validation sets for the training process, while the test set is reserved for performance evaluation.

3) LEVIR-Ship: The dataset comprises 3896 images captured from GF-1 and GF-6 satellites, with 2320, 788, and 788 images allocated to the training, validation, and test sets, respectively. The image size is 512 × 512. There are over 3000 annotated tiny ship objects. Most objects are smaller than 20 × 20 pixels, primarily clustering around 10 × 10 pixels. For this article, only the training set is utilized for training purposes, while the test set is reserved for assessing the final accuracy.

B. Training Setting

During the pretraining of the proposed backbone RepViT-TD in this article, we employed the timm codebase and utilized two NVIDIA RTX A6000 GPUs for training. The standard image size of 224 × 224 is adopted for training. The batch size is set to 256 on per GPU. The initial learning rate was set to 0.001, and a total of 300 epochs were trained. Additionally, various data augmentation techniques were utilized during training, such as color jitter and RandAugment.

Moreover, all tiny object detection experiments were conducted on a single NVIDIA RTX A6000 GPU. The model code is developed based on MMDetection [77], a PyTorch-based [76] object detection library. Training is performed using the stochastic gradient descent optimizer with a momentum of 0.9, a weight decay of 0.0001, and a batch size of 2. The initial learning rate is set to 0.001. A constant warm-up strategy with a ratio of 0.33 is applied for the first 10 000 iterations.

C. Evaluation Metrics

In this article, the different methods are evaluated using the widely used metric AP. The AP can be defined as

$$\text{AP} = \int_0^1 P(r)dr \tag{11}$$

where r indicates the recall and $P(r)$ denotes the precision. The AP is calculated by determining the area under the precision-recall curve. More precisely, AP50 refers to the IoU threshold of 0.5 used to define true positives, while AP75 denotes the IoU threshold of 0.75 for defining true positives. AP represents the average value computed across the range from AP50 to AP95, with IoU intervals set at 0.05.

For the AI-TODv2 dataset, we not only use AP, AP50, and AP75 for evaluation but also employ AP_{vt}, AP_{t}, AP_{s}, and AP_{m} to evaluate the detection accuracy of very tiny, tiny, small, and medium objects. These metrics are detailed in [22]. For the LEVIR-Ship dataset, only the AP50 is used to evaluate the detection accuracy.

D. Experimental Results

In this section, the efficacy of the proposed method for detecting tiny objects will be demonstrated. Several experiments will be presented, covering the effectiveness of the proposed backbone RepViT-TD, an ablation study of feature channels, comparisons with other methods, and the results on the LEVIR-Ship dataset.

1) Effectiveness of RepViT-TD: To validate the effectiveness of the proposed RepViT-TD backbone for tiny object detection, we conducted experiments using several detectors on the AI-TODv2 dataset. The detectors used in the experiments include Faster R-CNN [28], ATSS [60], FCOS [29], RetinaNet [30], and feature-and-spatial aligned network (FSANet) [24]. The detectors used in the experiments include both anchor-based and anchor-free methods, as well as a detector specifically designed for tiny object detection. Additionally, the compared backbones consist of the classic ResNet50 [31], the lightweight representative EfficientFormerV2 [71], and the baseline model RepViT [68]. These selections can provide a comprehensive evaluation of the performance advantages of RepViT-TD. The comparison results are shown in Table VI. From the table, it can be observed that when the backbone is replaced from ResNet50 to RepViT-TD, both the FLOPs and Params decrease significantly. For instance, in Faster R-CNN, the Params decrease from 41.66 M to 21.17 M, and the FLOPs decrease from 134.42 to 90.75 G. However, the detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
VALIDATION OF THE EFFECTIVENESS OF REPVIT-TD. ALL METHODS WERE TRAINED FOR 12 EPOCHS ON THE AI-TODV2 DATASET

Method	Backbone	Params(M)	FLOPs(G)	AP(%)	AP50(%)	AP75(%)	APvt(%)	APt(%)	APs(%)	APm(%)
Faster R-CNN[28]	ResNet50	41.16	134.42	12.8	29.9	9.4	0.0	9.2	24.6	37.0
	EfficientFormerV2	22.7	85.9	8.4	20.2	5.4	0.0	6.0	16.7	22.4
	RepViT	21.76	88.35	10.8	26.3	6.9	0.0	8.4	20.9	32.4
	RepViT-TD	21.17	90.75	13.6	31.2	9.9	0.0	10.3	25.7	31.0
ATSS[60]	ResNet50	32.07	128.38	13.0	31.0	8.7	2.3	11.2	18.0	29.9
	EfficientFormerV2	13.51	79.93	12.6	32.2	7.4	2.9	10.2	17.9	29.7
	RepViT	12.56	82.22	13.0	32.8	8.1	2.5	11.3	18.0	29.9
	RepViT-TD	11.97	84.46	13.3	32.2	8.6	3.5	11.4	18.7	31.8
FCOS[29]	ResNet50	31.85	123.16	15.0	37.5	9.1	3.2	13.4	20.3	31.5
	EfficientFormerV2	13.5	79.91	11.0	29.1	5.8	2.1	10.2	15.0	24.0
	RepViT	12.56	82.20	13.5	34.8	7.9	3.0	12.5	18.8	28.1
	RepViT-TD	10.79	83.27	15.6	39.2	9.4	3.3	14.0	21.4	33.3
RetinaNet[30]	ResNet50	36.25	129.75	8.9	24.2	4.6	2.7	8.4	13.1	20.4
	EfficientFormerV2	13.65	82.82	5.9	17.7	2.4	1.4	5.4	9.6	17.2
	RepViT	13.07	85.17	8.6	24.2	4.2	1.9	7.3	13.0	21.9
	RepViT-TD	12.87	87.46	8.9	25.6	4.0	2.5	7.8	13.8	22.5
FSANet[24]	ResNet50	30.7	109.71	17.6	45.0	10.5	5.4	15.8	22.9	33.8
	EfficientFormerV2	12.42	63.67	16.2	41.6	9.4	5.1	14.3	21.3	31.4
	RepViT	11.44	65.73	17.2	44.0	10.2	4.5	14.9	22.7	33.1
	RepViT-TD	10.83	68.20	17.7	45.2	10.7	5.2	15.7	24.2	33.8

TABLE VII
ABLATION EXPERIMENTS ON DIFFERENT FEATURE CHANNELS. ALL METHODS WERE TRAINED FOR 12 EPOCHS ON THE AI-TODV2 DATASET

Method	Feature channels	Params(M)	FLOPs(G)	AP(%)	AP50(%)	AP75(%)	AP_{vt}(%)	AP_t(%)	AP_s(%)	AP_m(%)
LTDNet	256	6.70	53.92	15.3	40.0	8.8	3.6	13.8	21.0	31.3
	128	5.26	23.57	15.0	39.3	8.6	3.3	13.5	20.6	31.3
	96	4.76	19.04	14.9	39.1	8.3	2.8	13.7	20.1	31.1
	64	4.39	15.60	13.9	36.8	7.5	3.1	12.8	19.2	29.2
	48	4.25	14.43	13.6	36.5	7.1	2.8	12.2	18.6	29.2

accuracy for tiny object is very competitive. Notably, AP_{vt} of Faster R-CNN is 0.0%, primarily because the anchor sizes differ considerably from the ground-truth boxes due to their extremely small size. Compared to Efficientformerv2, RepViT-TD has similar Params and FLOPs, but their detection accuracy differs significantly. For example, on Faster R-CNN, using RepViT-TD as the backbone improved AP by 5.2% compared to using the Efficientformerv2 backbone, and AP improved by 4.6% on the FCOS detector. Finally, compared to the baseline RepViT, the Params and FLOPs of RepViT-TD and RepViT are almost equal, but there is a significant improvement in tiny object detection performance on different detectors. Specifically, the mAP increased by 2.8% on Faster R-CNN, 2.1% on FCOS, and 0.5% on FSANet. Therefore, it can be concluded that the proposed RepViT-TD demonstrates highly competitive performance in tiny object detection in remote sensing images while maintaining the lightweight of the model.

2) Ablation Study of Feature Channels: Due to the significant impact of feature channels on computational complexity in LTDNet, we conducted ablation experiments on different feature channels. Five sets of channels are selected, 48, 64, 96, 128, and 256. The experimental results on different feature channels are shown in Table VII. From the table, it can be observed that when the feature channels are 256, the FLOPs are 53.92G, while when the feature channels are 128, the FLOPs decrease to 23.57 G. As the number of channels decreases, the FLOPs significantly reduce. Therefore, selecting an appropriate number of feature channels is crucial for reducing the model complexity. Specifically, when the feature channels decrease from 256 to 96, there is not too much change in the detection accuracy for tiny objects. However, when the feature channels decrease to 64 or 48, there is a significant drop in detection accuracy. Therefore, considering the tradeoff between accuracy and complexity, we set the feature channels of the feature fusion and detection head in LTDNet to 96.

3) Compared With Other Methods: We compared our proposed algorithm, LTDNet, with a series of representative object detection methods. These include classical two-stage frameworks such as TridentNet [78], Faster R-CNN [28], Cascade R-CNN [49], and DetectoRS [51]; anchor-based and anchor-free one-stage detectors such as FCOS [29], ATSS [60], RetinaNet [30], RepPoints [44], FoveaBox [32], and Grid R-CNN [78]; as well as recent state-of-the-art methods specifically designed for tiny object detection, including DotD [20], Cascade R-CNN w/normalized Wasserstein distance and a new ranking-based assigning (NWD-RKA) [22], DNTR [25], FSANet [24], ORFENet [26], and ESG_TODNet [27]. These comparison methods can comprehensively reflect the competitiveness of LTDNet. We followed the official implementations or widely adopted configurations for all comparison methods. Specifically, DNTR, ORFENet, and ESG_TODNet were trained for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

ALL-CLASS PERFORMANCE OF THE PROPOSED LTDNET AND OTHER STATE-OF-THE-ART METHODS ON THE AI-TOD-V2 TEST SET. BOLD FONTS INDICATE THE BEST PERFORMANCE. LTDNET WAS TRAINED FOR 36 EPOCHS. IN THE LAST ROW, $*$ DENOTES THE PROPOSED METHOD THAT WAS TRAINED FOR 36 EPOCHS WHEN THE P2 FEATURE MAP IS USED. THE PROPOSED METHOD CAN ACHIEVE THE BEST RESULTS IN TERMS OF FLOPS AND PARAMS

Method	FLOPs(G)	Params(M)	AP(%)	AP50(%)	AP75(%)	AP_{vt}(%)	AP_t(%)	AP_s(%)	AP_m(%)
Two-stage:									
TridentNet [78]	789.81	33.57	10.1	24.5	6.7	0.1	6.3	19.8	31.9
Faster R-CNN w/ ATSS [28][60]	134.42	41.16	12.8	29.6	9.2	0.0	9.2	24.9	36.6
Faster R-CNN -R50 [28]	134.42	41.16	12.8	29.9	9.4	0.0	9.2	24.6	37.0
Faster R-CNN -R101 [28]	182.33	60.52	13.1	30.7	9.2	0.0	9.7	24.6	35.5
Faster R-CNN -HRNet-w32 [28]	183.37	46.91	14.5	32.8	10.6	0.1	11.1	27.4	37.8
Cascade-R-CNN [49]	162.21	68.95	15.1	34.2	11.2	0.1	11.5	26.7	38.5
DetectoRS [51]	159.68	123.22	16.1	35.5	12.5	0.1	12.6	28.3	**40.0**
DotD [20]	134.42	41.16	20.4	51.4	12.3	8.5	21.1	24.6	30.4
Cascade-R-CNN w/ NWD-RKA [22]	162.21	68.95	22.2	52.5	15.1	7.8	21.8	28.0	37.2
DNTR [25]	298.13	128.37	23.1	51.9	16.8	9.7	22.7	27.2	36.1
One-stage:									
FCOS [29]	123.16	31.85	15.0	37.5	9.1	3.2	13.4	20.3	31.5
ATSS [60]	128.38	32.07	13.0	31.0	8.7	2.3	11.2	18.0	29.9
RetinaNet [30]	129.75	36.25	8.9	24.2	4.6	2.7	8.4	13.1	20.4
RepPoints [46]	118.71	36.60	9.3	23.6	5.4	2.8	10.0	12.3	18.9
FoveaBox [79]	129.01	36.19	11.3	28.1	7.4	1.4	8.6	17.8	32.2
Grid R-CNN [81]	247.94	64.32	14.3	31.1	11.0	0.1	11.0	25.7	36.7
FSANet [24]	109.71	30.70	17.6	45.0	10.5	5.4	15.8	22.9	33.8
ORFENet [26]	373.34	32.77	**24.8**	**55.4**	**18.2**	**9.7**	**24.4**	28.7	35.1
ESG_TODNet [27]	387.22	32.88	24.6	55.1	18.1	9.5	24.0	**29.4**	35.6
LTDNet(ours)	**19.04**	**4.76**	18.9	46.7	12.1	5.3	17.9	24.2	34.3
LTDNet*(ours)	**38.19**	**4.85**	23.0	54.6	15.5	8.9	23.6	27.2	33.1

TABLE IX

CLASS-WISE AP OF THE PROPOSED LTDNET AND OTHER EXCELLENT METHODS ON THE AI-TOD-V2 TEST SET. BOLD FONTS INDICATE THE BEST PERFORMANCE. IN THIS TABLE, AP IS THE EVALUATION METRIC OF EACH CLASS. THE SHORT NAME FOR CATEGORIES IS DEFINED AS AIRPLANE (AI), BRIDGE (BR), STORAGE TANK (ST), SHIP (SH), SWIMMING POOL (SP), VEHICLE (VE), PERSON (PE), AND WIND-MILL (WM). LTDNET WAS TRAINED FOR 36 EPOCHS. IN THE LAST ROW, $*$ DENOTES THE PROPOSED METHOD THAT WAS TRAINED FOR 36 EPOCHS WHEN THE P2 FEATURE MAP IS USED. SPECIFICALLY, THE PROPOSED CAN ACHIEVE THE BEST RESULTS IN TERMS OF FLOPS AND PARAMS AND CAN ACHIEVE THE BEST PERFORMANCE IN THE CLASSES OF SP

Method	FLOPs(G)	Params(M)	AI	BR	ST	SH	SP	VE	PE	WM
Two-stage:										
TridentNet [78]	789.81	33.57	19.3	0.1	17.2	16.2	12.4	12.5	3.4	0.0
Faster R-CNN w/ ATSS [28][60]	134.42	41.16	24.7	5.5	20.5	19.8	12.0	15.1	4.8	0.0
Faster R-CNN -R50 [28]	134.42	41.16	19.7	4.8	19.0	19.9	3.7	14.4	4.8	0.0
Faster R-CNN -R101 [28]	182.33	60.52	25.3	8.5	19.4	19.9	12.5	14.6	4.5	0.0
Faster R-CNN -HRNet-w32 [28]	183.37	46.91	27.9	9.5	21.5	21.4	13.0	16.7	5.9	0.0
Cascade-R-CNN [49]	162.21	68.95	26.2	9.6	24.0	24.3	13.2	17.5	5.8	0.1
DetectoRS [51]	159.68	123.22	**28.5**	11.7	23.2	26.4	14.9	17.6	6.5	0.2
DotD [20]	134.42	41.16	18.7	17.5	34.7	37.0	12.4	25.4	10.3	7.4
Cascade-R-CNN w/ NWD-RKA [22]	162.21	68.95	28.5	17.5	**36.9**	38.3	13.7	26.6	10.4	5.7
DNTR [25]	298.13	128.37	29.5	14.7	34.9	50.8	14.6	25.3	10.4	4.6
One-stage:										
FCOS [29]	123.16	31.85	14.7	12.2	24.4	31.3	12.6	17.0	5.2	2.2
ATSS [60]	128.38	32.07	15.4	11.7	20.0	27.6	9.4	14.8	4.7	0.0
RetinaNet [30]	129.75	36.25	1.3	11.8	14.3	23.6	5.8	11.4	2.3	0.5
RepPoints [46]	118.71	36.6	0.0	0.1	22.5	28.8	0.2	18.3	4.1	0.0
FoveaBox [79]	129.01	36.19	15.6	3.3	21.1	20.8	9.7	16.3	4.0	0.0
Grid R-CNN [81]	247.94	64.32	24.5	11.7	20.9	23.5	12.1	16.1	5.1	0.4
FSANet [24]	109.71	30.7	19.2	16.0	28.3	33.0	12.9	20.4	6.0	5.3
ORFENet [26]	373.34	32.77	26.0	**21.1**	35.8	50.6	17.1	27.8	11.0	**8.7**
ESG_TODNet [27]	387.22	32.88	26.6	20.5	35.5	**50.7**	16.5	27.8	**11.2**	8.0
LTDNet(ours)	**19.04**	**4.76**	24.1	16.2	26.1	36.8	16.9	20.1	6.1	5.4
LTDNet*(ours)	**38.19**	**4.85**	23.3	19.9	34.4	44.8	**17.2**	27.4	9.6	7.3

36 epochs with a batch size of 2, while FSANet was trained for 12 epochs with a batch size of 4. The remaining methods were trained for 12 epochs with a batch size of 2. The input images were resized to 800 $\times$ 800 for both training and evaluation. The results are shown in Tables VIII and IX. From Table VIII, it can be seen that LTDNet achieves the best results in terms of FLOPs and Params, with 19.04 G FLOPs and 4.76 M Params, respectively. When the P2 feature map is adopted,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

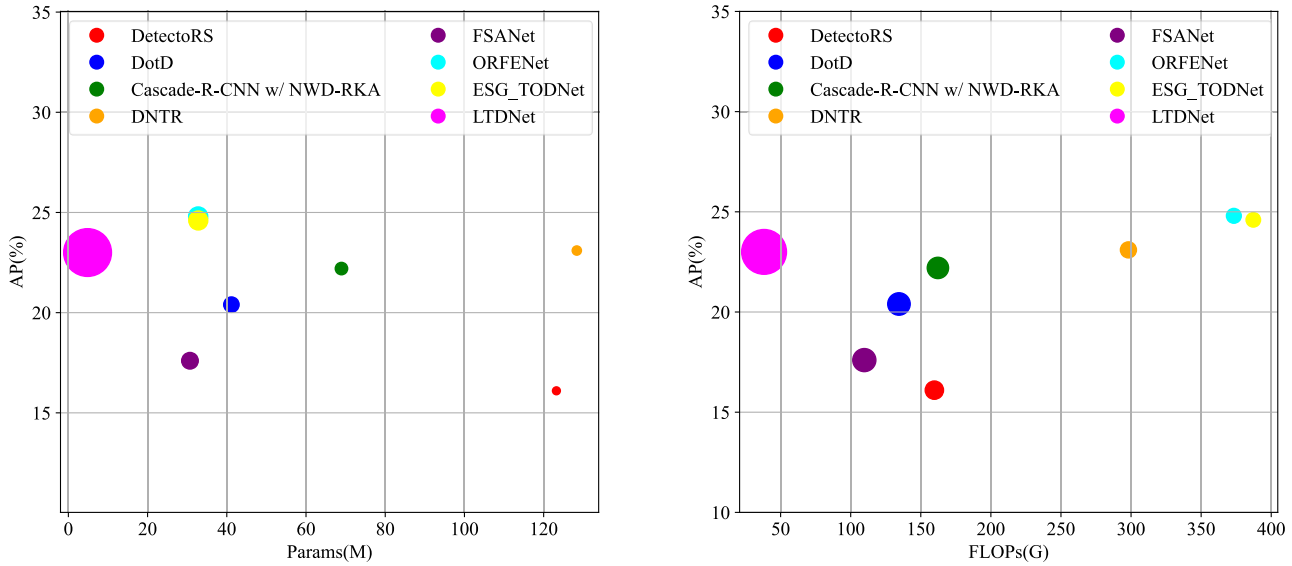


Fig. 6. Visualization of AP, Params, and FLOPs for different methods. The left figure illustrates the relationship between AP and Params for different methods, while the right figure shows the relationship between AP and FLOPs. The size of the circles represents the ratio of AP to Params or FLOPs.

TABLE X

EXPERIMENTAL RESULTS OF ESG_TODNET AND ORFENET USING LTDNET AS THE BASELINEON THE AI-TODV2 DATASET. ALL METHODS WERE TRAINED FOR 36 EPOCHS. COMPARED WITH THE ORIGINAL LTDNET, ESGTODNET AND ORFENET CAN ACHIEVE BETTER ACCURACY FOR TINY OBJECTS

Method	FLOPs(G)	Params(M)	AP(%)	AP50(%)	AP75(%)	APvt(%)	APt(%)	APs(%)	APm(%)
LTDNet	38.19	4.85	23.0	54.6	15.5	8.9	23.6	27.2	33.1
ESG_TODNet [27]	45.73	5.27	23.9	55.6	16.8	9.6	24.2	27.7	34.1
ORFENet [26]	45.39	5.09	24.1	56.2	16.4	10.1	24.6	28.9	35.4

the FLOPs and Params are 38.19 G and 4.85 M, respectively. Compared to other methods, LTDNet significantly reduces both FLOPs and Params. Importantly, despite lower FLOPs and Params, LTDNet achieves competitive detection accuracy for tiny objects. As shown in Table VIII, the highest AP value is achieved by ORFENet, with an AP of 24.8%, while LTDNet achieves an AP of 23%. Compared to ORFENet, LTDNet requires approximately 10% of ORFENet's FLOPs and 14.8% of its Params while achieving 92.7% of its AP. Furthermore, the difference in AP50 is even smaller, with ORFENet at 55.4% and LTDNet at 54.6%, reaching 98.6% of ORFENet. LTDNet demonstrates competitive performance while maintaining significantly lower Params and FLOPs. This demonstrates that the structure of LTDNet is well-suited for tiny object detection. Additionally, from Table IX, LTDNet shows competitive accuracy across different categories, achieving the highest detection accuracy on the "swimming pool" category. The main reason is that the proposed lightweight model, due to its limited capacity, tends to prioritize learning easily distinguishable features. The category of "swimming pool" aligns well with clear and consistent visual features. Therefore, it exhibits better detection performance. This further confirms the effectiveness of the proposed LTDNet.

Moreover, the AP, Params, and FLOPs of different methods have been visualized, as shown in Fig. 6. The figures illustrate the relationship between Params and AP and between FLOPs and AP, respectively. The circle size of each method represents the ratio of AP to Params and AP to FLOPs. These visualizations provide a comprehensive view of the performance of different algorithms for tiny object detection. Specifically, points closer to the top-left corner indicate higher Params efficiency and higher FLOPs efficiency. Additionally, larger sizes represent better overall performance. From the figures, it is evident that the proposed method is closer to the top-left corner and has the largest size compared to other methods. Therefore, the proposed method is more effective. In addition, to further validate the real-world applicability of our method, we conducted inference speed evaluations on the Jetson AGX Orin edge device. The results show that our model

TABLE XI

PERFORMANCE OF THE PROPOSED LTDNET AND SEVERAL EXCELLENT METHODS ON THE LEVIR-SHIP DATASET. THE PROPOSED LTDNET WITHOUT P2 FEATURE MAP CAN ACHIEVE THE RESULTS OF 84.0% AP50

Method	Params(M)	FLOPs(G)	AP50(%)
YOLOv3 [41]	61.52	99.2	69.9
SSD [38]	24.39	175.2	52.6
YOLOv5s [83]	7.05	10.4	75.6
EfficientDet-D2 [81]	8.01	20.0	71.3
Faster R-CNN(VGG16) [28]	136.7	299.2	70.8
CenterNet [44]	191.24	584.6	77.7
HSFNet [82]	157.59	538.1	73.6
DRENet [75]	4.79	8.3	82.4
ORFENet [26]	32.77	152.54	**84.9**
LTDNet(Ours)	**4.75**	**7.75**	84.0

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

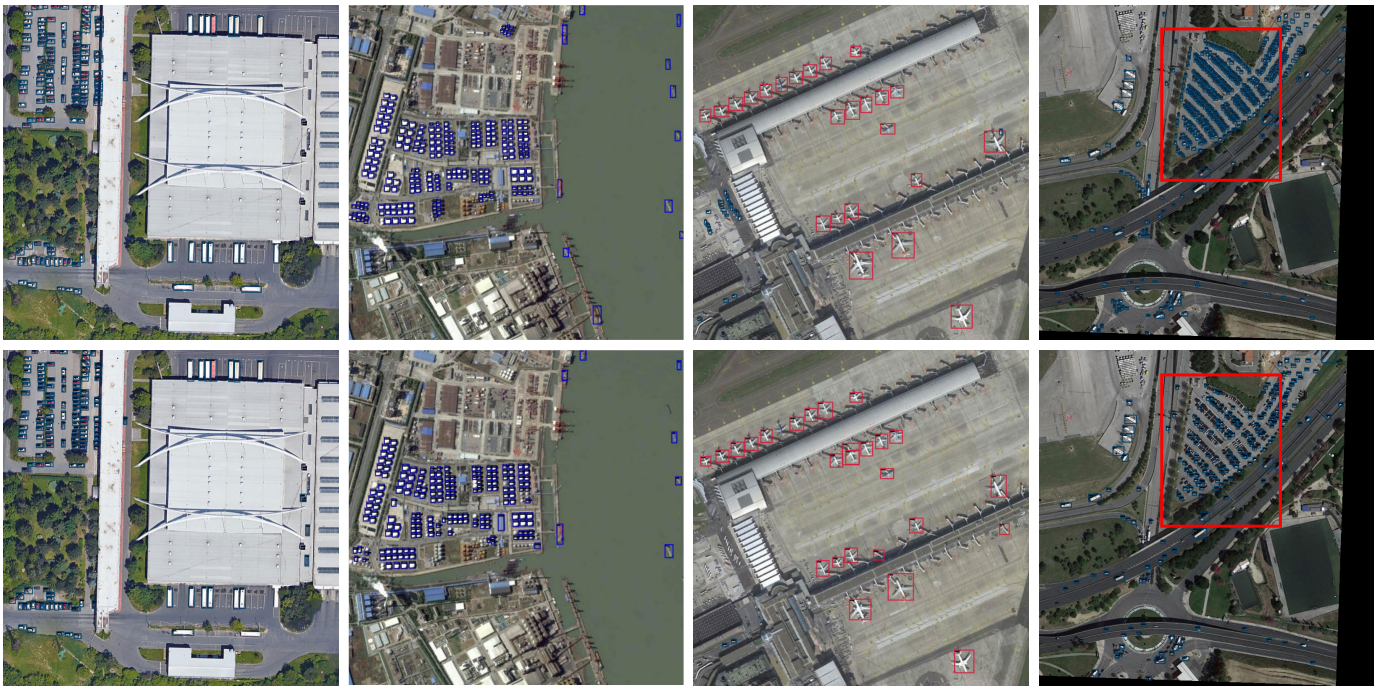


Fig. 7. Visual results of the proposed method on the AI-TODv2 dataset. The first row corresponds to the ground truth annotations, and the second row displays the predicted results. As can be seen from the figure, the proposed method can achieve pretty good results for several kinds of objects such as vehicles, storage tanks, and ships.

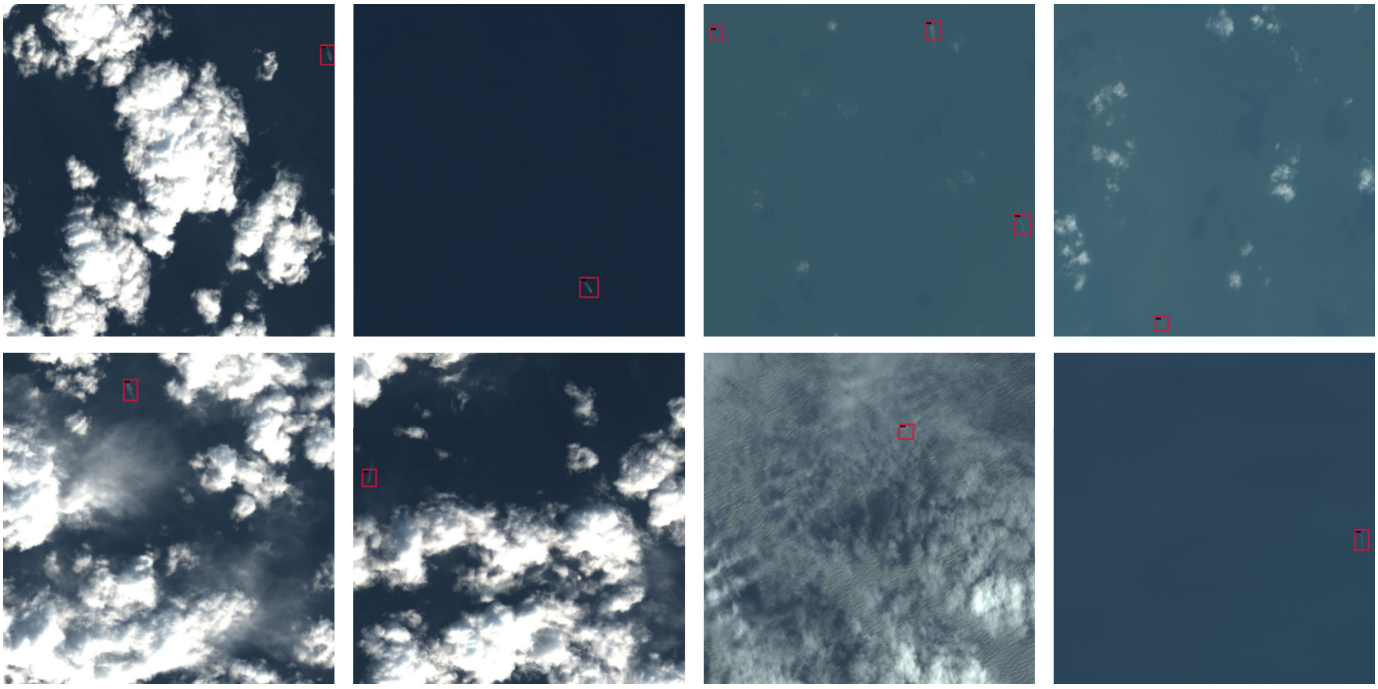


Fig. 8. Detection results of the proposed LTDNet on the LEVIR-Ship dataset. The proposed method can achieve good detection results in different situations with the lowest FLOPs and Params.

achieves 27 FPS with a resolution of 800 × 800, highlighting its strong potential in resource-constrained environments.

4) LTDNet as Baseline: In order to validate that the proposed method can effectively serve as a baseline for other methods, we conducted experiments using LTDNet as the baseline on ORFENet [26] and ESG_TODNet [27]. The experimental results are shown in Table X. From the table, it can be observed that when using LTDNet as the baseline for tiny object detection, both ESG_TODNet and ORFENet show improved detection accuracy. Specifically, for ESG_TODNet, the AP increases from the baseline of 23.0%–23.9%, AP50 increases from 54.6% to 55.6%, and AP75 increases from 15.5% to 16.8%. Similarly, for ORFENet, the AP increases from the baseline of 23.0% to 24.1%, AP50 increases from 54.6% to 56.2%, and AP75 increases from 15.5% to 16.4%. Additionally, both methods show improvements

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

in AP_{vt}, AP_t, AP_s, and AP_m. Moreover, compared to the original algorithms, FLOPs and Params of ESG_TODNet and ORFENet with LTDNet as the baseline are significantly lower. This demonstrates the importance of LTDNet in the field of tiny object detection in remote sensing images.

5) Several Visualization Results: The visualization results of the proposed LTDNet on the AI-TODv2 dataset are shown in Fig. 7, where all detection results with confidence greater than 0.3 are displayed. It is evident from the figure that LTDNet performs well in the categories of vehicles, storage tanks, and ships. Moreover, whether in scenes containing only a single class of airplane or in multiclass object scenes, such as ships and storage tanks, LTDNet achieves good detection results. This demonstrates the effectiveness of the proposed LTDNet for detecting tiny objects in remote sensing images. However, for objects that are extremely small and highly clustered, as shown in the red box in the last column, the detection results are not yet ideal. Therefore, more effective methods are still needed to address such scenarios. To this end, our future work will explore advanced modeling strategies, such as instance-level feature refinement, aiming to better handle object clustering and further enhance the robustness of tiny object detection.

6) Detection Results on LEVIR-Ship Dataset: To further validate the effectiveness of the proposed LTDNet, we conducted experiments on the LEVIR-Ship dataset and compared it with other state-of-the-art methods. The comparison results are shown in Table XI. From the table, it can be seen that LTDNet achieves the lowest FLOPs and Params, with a higher detection accuracy of 84.0% AP50. Compared to ORFENet, a method specifically designed for tiny object detection, LTDNet reduces FLOPs by 95% and Params by 85%. In this case, LTDNe can achieve a detection accuracy very close to that of ORFENet. Compared to the lightweight method DRENet, LTDNet has fewer Params and lower FLOPs but achieves higher detection accuracy. Therefore, this demonstrates the effectiveness of the proposed LTDNet. In addition, the detection results of the proposed LTDNet on the LEVIR-Ship dataset are shown in Fig. 8. From the figure, it can be seen that the proposed LTDNet can achieve relatively good detection results in different situations, including calm sea surfaces and thick clouds. Therefore, the proposed method is effective.

V. Conclusion

In this article, we first explore the impact of the complexity of different components in deep learning-based object detection models on the detection accuracy of tiny objects in remote sensing images. Subsequently, we propose a lightweight backbone RepViT-TD specifically designed for tiny object detection. This lightweight backbone can achieve excellent detection accuracy while maintaining lower FLOPs and Params. Additionally, we integrate the DCN into the detection head and optimize the channel dimension, effectively balancing detection accuracy and complexity. By integrating these improvements, we propose LTDNet, a lightweight network for tiny object detection, significantly reducing computational complexity and parameters while maintaining competitive accuracy. Extensive experiments have proven the effectiveness of the proposed method. Furthermore, its superior performance and efficiency make it well-suited for critical real-world remote sensing tasks, including military reconnaissance and maritime illegal fishing detection, thereby offering robust support for practical deployment in mission-critical scenarios. In the future, further research can be conducted on the lightweight problem of tiny object detection algorithms from the perspective of hardware–software co-design, in order to enhance the applicability. Specifically, this can be achieved by designing hardware-friendly neural network architectures tailored for embedded hardware (such as field-programmable gate array (FPGA) or AI chips), thereby improving the efficiency.

References

[1] Z. Lv, M. Zhang, W. Sun, J. A. Benediktsson, T. Lei, and N. Falco, "Spatial-contextual information utilization framework for land cover change detection with hyperspectral remote sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411911.

[2] Z. Lv, T. Yang, P. Zhong, W. Sun, J. A. Benediktsson, and J. Li, "Hierarchical feature fusion triple network for change detection with bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4401512.

[3] Z. Lv, P. Zhang, W. Sun, J. A. Benediktsson, and T. Lei, "Novel land-cover classification approach with nonparametric sample augmentation for hyperspectral remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407613.

[4] Y. Ding et al., "SLCGC: A lightweight self-supervised low-pass contrastive graph clustering network for hyperspectral images," 2025, *arXiv:2502.03497*.

[5] Y. Ding et al., "Adaptive homophily clustering: Structure homophily graph learning with adaptive filter for hyperspectral image," 2025, *arXiv:2501.01595*.

[6] J. Han, D. Zhang, G. Cheng, L. Guo, and J. Ren, "Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3325–3337, Jun. 2015.

[7] Y. Li, Y. Zhang, X. Huang, and A. L. Yuille, "Deep networks under scene-level supervision for multi-class geospatial object detection from remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 182–196, Dec. 2018.

[8] T. Gao, L. Gu, H. Chen, and B. Zhou, "Domain adaptation support tensor machine: An extended STM for object recognition using cross-source heterogeneous remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4507521.

[9] X. Dong, Y. Qin, R. Fu, Y. Gao, S. Liu, and Y. Ye, "Remote sensing object detection based on gated context-aware module," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[10] Y. Ye et al., "An adaptive attention fusion mechanism convolutional network for object detection in remote sensing images," *Remote Sens.*, vol. 14, no. 3, p. 516, Jan. 2022.

[11] R. Kou et al., "Infrared small target segmentation networks: A survey," *Pattern Recognit.*, vol. 143, Nov. 2023, Art. no. 109788.

[12] C. Xia, S. Chen, R. Huang, J. Hu, and Z. Chen, "Separable spatial–temporal patch-tensor pair completion for infrared small target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5001620.

[13] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, vol. 1, Jun. 2005, pp. 886–893.

[14] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ra-manan, "Object detection with discriminatively trained part-based mod-els," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, Sep. 2010.

[15] P. F. Felzenszwalb, R. B. Girshick, and D. McAllester, "Cascade object detection with deformable part models," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, San Francisco, CA, USA, Jun. 2010, pp. 2241–2248.

[16] Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," 2019, *arXiv:1905.05055*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

[17] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.

[18] Y. Long, Y. Gong, Z. Xiao, and Q. Liu, "Accurate object localization in remote sensing images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2486–2498, May 2017.

[19] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[20] C. Xu, J. Wang, W. Yang, and L. Yu, "Dot distance for tiny object detection in aerial images," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Nashville, TN, USA, Jun. 2021, pp. 1192–1201.

[21] J. Wang, W. Yang, H. Guo, R. Zhang, and G.-S. Xia, "Tiny object detection in aerial images," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 3791–3798.

[22] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G.-S. Xia, "Detecting tiny objects in aerial images: A normalized Wasserstein distance and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 79–93, Aug. 2022.

[23] C. Xu, J. Wang, W. Yang, H. Yu, L. Yu, and G. S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 536–543.

[24] J. Wu, Z. Pan, B. Lei, and Y. Hu, "FSANet: Feature-and-spatial-aligned network for tiny object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5630717.

[25] H.-I. Liu, Y.-W. Tseng, K.-C. Chang, P.-J. Wang, H.-H. Shuai, and W.-H. Cheng, "A DeNoising FPN with transformer R-CNN for tiny object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704415.

[26] D. Liu, J. Zhang, Y. Qi, Y. Wu, and Y. Zhang, "Tiny object detection in remote sensing images based on object reconstruction and multiple receptive field adaptive feature enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5616213.

[27] D. Liu, J. Zhang, Y. Qi, Y. Wu, and Y. Zhang, "A tiny object detection method based on explicit semantic guidance for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[29] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.

[30] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[31] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.

[32] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis., Aug.*, Jan. 2020, pp. 213–229.

[33] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. IEEE Conf. Int. Conf. Learn. Represent.*, Apr. 2021.

[34] D. Meng et al., "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3631–3640.

[35] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query DeNoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13609–13617.

[36] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," in *Proc. IEEE Conf. Int. Conf. Learn. Represent.*, Apr. 2022.

[37] H. Zhang et al., "DINO: DETR with improved denoising anchor boxes for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent.*, Apr. 2023.

[38] W. Liu et al., "SSD: Single shot MultiBox detector," in *Proc. Eur. Conf., Comput. Vis.*, Jan. 2016, pp. 21–37.

[39] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[40] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6517–6525.

[41] A. Farhadi and J. Redmon, "YOLOv3: An incremental improvement," in *Proc. Comput. Vis. Pattern Recognit.*, vol. 1804. Berlin, Germany: Springer, 2018, pp. 1–6.

[42] C.-Y. Wang, A. Bochkovskiy, and H. M. Liao, "Scaled-YOLOv4: Scaling cross stage partial network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13024–13033.

[43] H. Law and J. Deng, "CornerNet: Detecting objects as paired key-points," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 734–750.

[44] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[45] X. Zhou, J. Zhuo, and P. Krähenbühl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.

[46] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9656–9665.

[47] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Region-based convolutional networks for accurate object detection and segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 1, pp. 142–158, Jan. 2016.

[48] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[49] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6154–6162.

[50] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[51] S. Qiao, L.-C. Chen, and A. Yuille, "DetectoRS: Detecting objects with recursive feature pyramid and switchable atrous convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10208–10219.

[52] M. Kisantal, Z. Wojna, J. Murawski, J. Naruniec, and K. Cho, "Augmentation for small object detection," 2019, *arXiv:1902.07296*.

[53] C. Chen et al., "RRNet: A hybrid detector for object detection in drone-captured images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 100–108.

[54] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[55] P.-Y. Chen, J.-W. Hsieh, C.-Y. Wang, H.-Y. M. Liao, and M. Gochoo, "Residual bi-fusion feature pyramid network for accurate single-shot object detection," 2019, *arXiv:1911.12051*.

[56] Y. Luo et al., "CE-FPN: Enhancing channel information for object detection," *Multimedia Tools Appl.*, vol. 81, no. 21, pp. 30685–30704, 2021.

[57] M. Haris, G. Shakhnarovich, and N. Ukita, "Task-driven super resolution: Object detection in low-resolution images," in *Proc. Commun. Comput. Inf. Sci., Neural Inf. Process.*, Jan. 2018, pp. 387–395.

[58] J. Li, X. Liang, Y. Wei, T. Xu, J. Feng, and S. Yan, "Perceptual generative adversarial networks for small object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 1951–1959.

[59] Y. Bai, Y. Zhang, M. Ding, and B. Ghanem, "SOD-MTGAN: Small object detection via multi-task generative adversarial network," in *Proc. Comput. Vis. (ECCV)*, Jan. 2018, pp. 210–226.

[60] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9756–9765.

[61] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[62] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[63] X. Ding, X. Zhang, N. Ma, J. Han, G.-G. Ding, and J. Sun., "RepVGG: Making VGG-style ConvNets great again," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13733–13742.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.

[64] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.

[65] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet v2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 116–131.

[66] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 6848–6856.

[67] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn. (ICML)*, Long Beach, CA, USA, Jan. 2019, pp. 6105–6114.

[68] A. Wang, H. Chen, Z. Lin, J. Han, and G. Ding, "RepViT: Revisiting mobile CNN from ViT perspective," 2023, *arXiv:2307.09283*.

[69] S. Mehta and M. Rastegari, "MobileViT: Light-weight, general-purpose, and mobile-friendly vision transformer," 2021, *arXiv:2110.02178*.

[70] Y. Li et al., "EfficientFormer: Vision transformers at mobilenet speed," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 12934–12949.

[71] Y. Li et al., "Rethinking vision transformers for MobileNet size and speed," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16843–16854.

[72] J. Ning, H. Guan, and M. Spratling, "Rethinking the backbone architecture for tiny object detection," 2023, *arXiv:2303.11267*.

[73] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 9300–9308.

[74] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[75] J. Chen, K. Chen, H. Chen, Z. Zou, and Z. Shi, "A degraded reconstruction enhancement-based method for tiny ship detection in remote sensing images with a new large-scale dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625014.

[76] A. Paszke et al., "PyTorch: An imperative style, high performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 8026–8037.

[77] K. Chen et al., "MMDetection: Open MMLab detection toolbox and benchmark," 2019, *arXiv:1906.07155*.

[78] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6053–6062.

[79] T. Kong, F. Sun, H. Liu, Y. Jiang, L. Li, and J. Shi, "FoveaBox: Beyound anchor-based object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 7389–7398, 2020.

[80] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 7355–7364.

[81] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.

[82] Q. Li, L. Mou, Q. Liu, Y. Wang, and X. X. Zhu, "HSF-net: Multiscale deep feature embedding for ship detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7147–7161, Dec. 2018.

[83] G. Jocher et al., "Ultralytics/YOLOv5: V5.0-YOLOv5-P6 1280 models AWS supervise. Ly and YouTube integrations," Zenodo, Geneva, Switzerland, Tech. Rep. Version V5.0, Apr. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:37 UTC from IEEE Xplore. Restrictions apply.